

MCBB 3.0 AWS LAB-POC

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1. AWS Cloud WAN Terminology & Concepts

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Quick reference for key AWS Cloud WAN terms and policy constructs used throughout this lab build.

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1.1 Core Components

Term	Definition
Core Network	The global C-WAN resource that spans all regions; contains the policy document, segments, and attachments.
Core Network Edge (CNE)	A regional presence of the core network. Each AWS region in your policy gets a CNE that handles routing and forwarding for that region.
Segment	An isolated routing domain within C-WAN, similar to a VRF. Attachments in the same segment can communicate; different segments cannot unless explicitly peered.
Segment Peering	Allows route exchange between two segments within the same CNE. Used to enable Shared Services to reach both Prod and Non-Prod.
Attachment	A connection between a resource (VPC, DXGW, S2S VPN, TGW, or Connect) and the core network. Each attachment belongs to exactly one segment.
Attachment Policy	Rules that automatically map new attachments to segments based on tag conditions. Evaluated in order; first match wins.
Network Route	A route entry in a segment's route table at a specific CNE, showing destination prefix, attachment, and route type (propagated or static).

1.2 Policy Actions & Routing Constructs

Term	Definition
Segment Actions	Policy rules that define how traffic is routed between segments and attachments. Includes share-with, send-to, and send-via actions.
Send To	Forwards traffic destined for a specific segment to an intermediary attachment (e.g., Inspection VPC) before delivery. Used for single-hop inspection.
Send Via	Inserts a middlebox (e.g., firewall) into the traffic path between two segments. Supports both single-hop and dual-hop inspection patterns.
Single Hop	Traffic passes through one intermediary attachment before reaching destination. Example: Prod VPC Inspection VPC On-Prem via DXGW.
Dual Hop	Traffic passes through two intermediary attachments before reaching destination. Example: Source VPC Inspection (region A) Inspection (region B) Dest VPC.
Share With	Shares routes from one segment with another without full peering. Provides one-directional or selective route sharing.
When (Condition)	A condition block in the attachment policy that matches on resource tags (e.g., tag:Environment = Prod) to determine segment placement.
Edge Override	Allows per-region overrides of the global segment/attachment policy. Used when a specific CNE needs different routing behavior.
Network Function Group	Allows you to automatically steer same-segment or cross-segment traffic using <i>network functions</i> deployed in VPCs or on-premises <i>network</i>

2. LAB Overview & Objectives

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2.1 Purpose

The MCBB 3.0 AWS Lab environment serves as a pre-production validation platform to prove out the Cloud WAN with Split DXGW architecture defined in the MCBB 3.0 production design document (v.25, Jan 2026). The lab mirrors the production topology at a reduced scale, enabling the team to validate design requirements (DRs), test failure scenarios, refine Terraform modules, and document operational runbooks before production deployment.

2.2 Scope

- AWS Cloud WAN (C-WAN) core network across three lab regions: us-east-1, ap-southeast-2, us-west-2
- C-WAN segmentation: Prod, Non-Prod, Shared Services (peered to Prod + Non-Prod), and IBB segments
- Cloud Network Edge (CNE) attachments per region
- Split DXGW architecture with regional DXGW instances
- Multiple VPC types per region: Prod, Non-Prod, Inspection, and Shared Services
- Inspection VPC for firewall deployment (AWS Network Firewall or third-party appliance)
- Direct Connect integration via Megaport virtual cross-connects (VXCs)
- BGP route propagation validation with community tagging
- Cross-region traffic flow validation over C-WAN backbone
- International Backbone (IBB) overlay design
- Service chaining through inspection VPCs for east-west and north-south traffic
- Infrastructure as Code (Terraform Enterprise) with GitLab CI/CD runners
- Network observability: interim CloudWatch/VPC Flow Logs, target state ThousandEyes + Splunk + Selector AI

2.3 Out of Scope

- Full-scale 100GE Cloud OnRamp capacity testing
- LATAM region deployment (Argentina, Panama, Mexico)
- Inter-CSP connectivity (Azure, GCP integration points)
- QoS enforcement at egress (DR 4.5 — not supported by CSP)
- Versa SD-WAN / SDR overlay — removed from design due to operational issues

2.4 Key Design Change: Versa Removal

The original MCBB 3.0 design specified Versa SDRs for the International Backbone (IBB) SD-WAN overlay. For AWS Cloud Inter/intra regional connectivity due to recent operational issues with Versa, the overlay between regions connectivity will be re-architected using cloud-native CNE (CloudWAN Network Edge):

- AWS Cloud WAN native intra/cross-region routing
- Public VIF over existing DX connections for IBB underlay (as referenced in production design doc)

2.5 Inspection VPC Architecture

Each region includes a dedicated Inspection VPC for centralized security appliances supporting east-west inspection (inter-VPC traffic routed through firewall before reaching destination), north-south inspection (traffic to on-prem/internet inspected before leaving the region), and service chaining via C-WAN routing policies (send-to / send-via actions).

Firewall options under evaluation:

- AWS Network Firewall — cloud-native, integrates with VPC route tables
 - Third-party appliance firewall (e.g., Palo Alto, CheckPoint, Fortinet) — deployed as EC2-based NVAs
-

3. LAB Topology & Network Diagrams

3.1 Lab Region Selection

Lab Region	AWS Region	VPCs Deployed
US East	us-east-1 (N. Virginia)	Prod, Non-Prod, Inspection, Shared Svc
APAC	ap-southeast-2 (Sydney)	Prod, Inspection
US West	us-west-2 (Oregon)	Prod, Non-Prod, Inspection
US East-2	us-east-2 (Ohio)	Prod, Non-Prod, Inspection

3.2 VPC Architecture Per Region

us-east-1 (N. Virginia)

VPC Name	CIDR	Purpose	C-WAN Segment
lab-prod-use1	10.10.184.128/25	Production workloads	Prod
lab-nonprod-use1	10.10.185.0/25	Non-production / dev / staging	Non-Prod
lab-inspection-use1	100.64.0.0/24	Firewall appliances (AWS NFW or 3rd party NVA)	Inspection attachment
lab-shared-use1	TBD	Shared services (DNS, logging, tooling)	Shared Services (peers to Prod + Non-Prod)

us-east-2 (Ohio)

VPC Name	CIDR	Purpose	C-WAN Segment
lab-prod-use2	10.10.186.0/25	Production workloads	Prod
lab-inspection-use2	100.64.1.0/24	Firewall appliances (AWS NFW or 3rd party NVA)	Inspection attachment
lab-nonprod-use2	10.10.186.128/25	Non-production / dev / staging	Non-Prod

us-west-2 (Oregon) (MnA/Client integration Simulation)

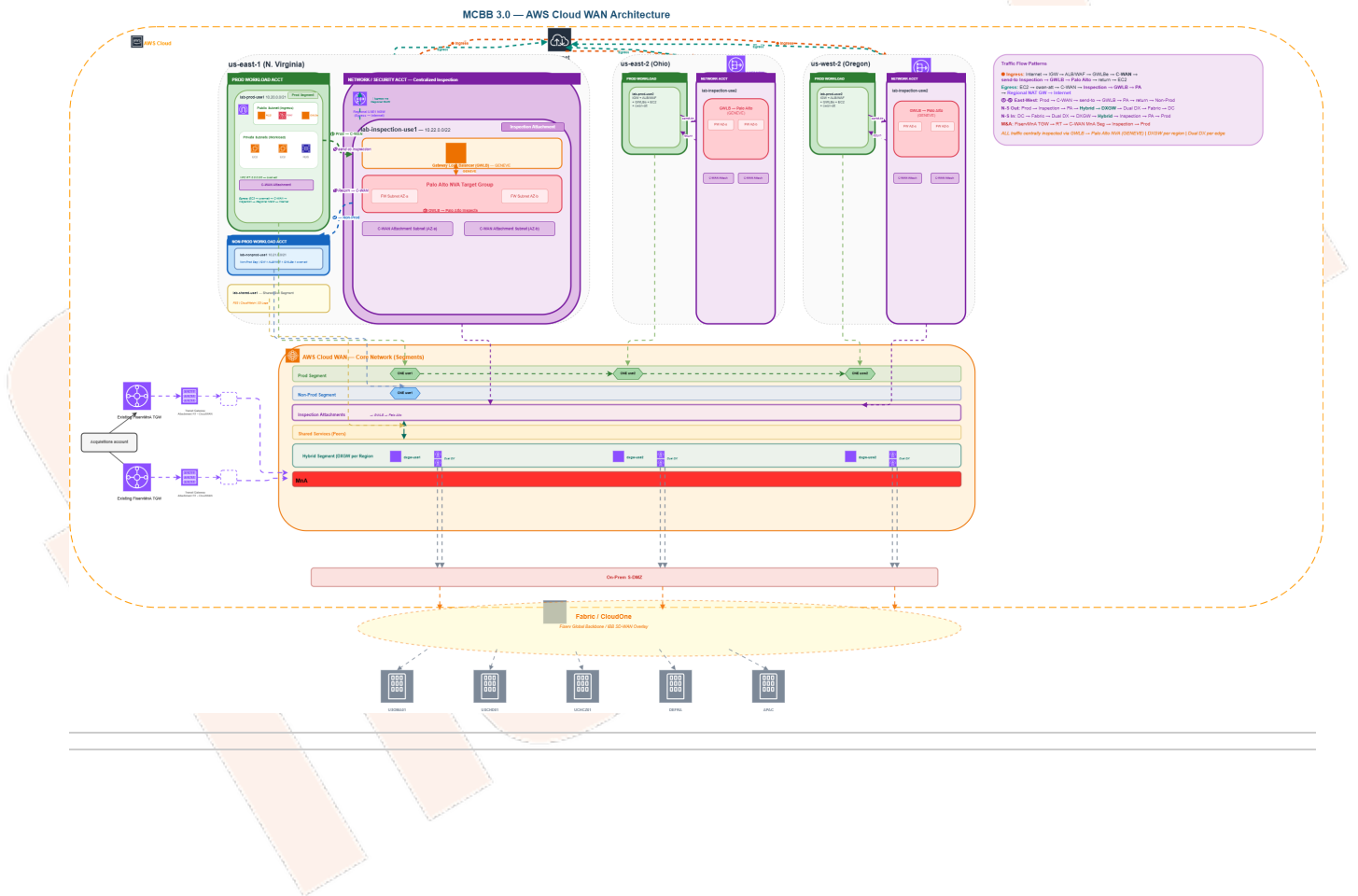
VPC Name	CIDR	Purpose	C-WAN Segment
lab-prod-usw2	10.10.187.0/25	Production workloads	Prod

lab-nonprod-usw2	10.10.187.128/25	Non-production / dev / staging	Non-Prod
lab-inspection-usw2	100.64.2.0/24	Firewall appliances (AWS NFW or 3rd party NVA)	Inspection attachment

Note: us-west-2 VPC CIDRs must be from a different address range than us-east-1 (10.20.x.x) and Sydney (10.30.x.x) to avoid overlap.

3.3 Low Level Lab Architecture Diagram - CloudWAN

A single comprehensive low-level diagram covering the full lab architecture: C-WAN core network with all 8 segments (Prod, Non-Prod, Shared Services, Hybrid, Clients & MnA), CNEs per region, VPC attachments with segment mapping, Inspection VPCs with firewall placement and service chaining flow (send-to/send-via), split DXGW instances with DX /Megaport connections, route propagation paths (CNE-to-CNE and CNE-to-DXGW), cross-region traffic flows, and IBB routing for international traffic, which will transverse via the CNE edges on AWS backbone.



5. Traffic Flow Description

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MCBB 3.0 — Traffic Flow Descriptions

AWS Cloud WAN Lab Architecture — Decentralized Ingress/Egress | Centralized Inspection (GWLBe + Palo Alto NVA) | DXGW per Region | Dual DX per Edge Connect

Flow Direction	Inspection?	C-WAN Policy	Key Component
Internet EC2	Yes — centralized	send-to Inspection	Per-region IGW + ALB/WAF + GWLBe
Egress EC2 Internet	Yes — centralized	send-to Inspection	Regional NAT GW (per Inspection VPC)
- East-West Prod Non-Prod	Yes — centralized	send-to Inspection	Segment isolation enforcement
N-S Outbound EC2 On-Prem	Yes — centralized	send-via Inspection	DXGW per region + Dual DX + Fabric
N-S Inbound On-Prem EC2	Yes — centralized	send-via Inspection	DXGW per region + Dual DX + Fabric
M&A Acquisitions Prod	Yes — centralized	MnA Inspection	FiservMnA TGW + TGW Attachment RT
Shared Any Seg	No — direct peering	Segment peering	R53, CloudWatch, S3 Logs

Ingress (Internet Workload e.g EC2)

Internet Regional IGW (per-region, e.g. IGW use1) Workload VPC Public Subnet ALB receives traffic WAF inspects L7 ALB forwards to GWLBe (in workload VPC) GWLBe routes via ingress routing table to C-WAN Attachment C-WAN Prod Segment C-WAN policy: send-to Inspection Segment Inspection VPC C-WAN Attachment Subnet GWLB (GENEVE encapsulation) Palo Alto NVA inspects (IPS/IDS, threat prevention, URL filtering) returns to GWLB C-WAN Attachment Subnet re-enters C-WAN Prod Segment back to Workload VPC EC2 in private subnet

Key: Even though each workload VPC has its own IGW and ALB/WAF (decentralized ingress), Cloud WAN is in the path. All traffic is centrally inspected via GWLB + Palo Alto before reaching EC2. The GWLBe in the workload VPC is what redirects ingress traffic into the C-WAN inspection path.

Egress (Workload EC2 Internet)

EC2 (private subnet) Workload VPC route table: 0.0.0.0/0 cwan-att **C-WAN Attachment** in workload VPC **C-WAN Prod Segment** C-WAN policy: **send-to Inspection Segment** **Inspection VPC C-WAN Attachment Subnet** **GWLB** (GENEVE) **Palo Alto NVA** inspects outbound (URL filtering, data exfiltration prevention) returns to **GWLB** **Inspection VPC FW Subnet** route table: 0.0.0.0/0 **Regional NAT GW** **Regional NAT GW** (e.g. **Regional USE1 NGW**) NAT translates private public IP **Regional IGW** **Internet**

Key: Egress does NOT use the workload VPC's NAT GW. Traffic exits through the **Regional NAT GW** associated with the **Inspection VPC** in the **Network/Security Account**. This ensures all outbound traffic is inspected by **Palo Alto** before leaving AWS. Each region has its own **Regional NGW** for egress.

- East-West (Prod VPC Non-Prod VPC)

EC2 in Prod VPC VPC route table: 0.0.0.0/0 cwan-att traffic enters **C-WAN Prod Segment**

C-WAN segment policy: **send-to Inspection Segment** (Prod cannot reach Non-Prod directly)

Inspection VPC C-WAN Attachment Subnet GWLB (GENEVE) Palo Alto NVA inspects (microsegmentation, inter-segment policy)

returns to **GWLB C-WAN Attachment Subnet** re-enters C-WAN routed to **Non-Prod Segment**

traffic delivered to **Non-Prod VPC EC2** in Non-Prod

Key: Segment isolation is enforced by C-WAN policy. Prod and Non-Prod are separate segments — they cannot communicate without traversing the Inspection segment. This is the "send-to" routing mode. Same pattern applies for any cross-segment communication.

North-South Outbound (Workload EC2 On-Prem DC)

EC2 (Prod VPC) VPC route table: on-prem prefixes (e.g. 10.0.0.0/8 cwan-att) **C-WAN Prod Segment** C-WAN policy: **send-via Inspection Segment** **Inspection VPC C-WAN Attachment Subnet** **GWLB** (GENEVE) **Palo Alto NVA** inspects returns to **GWLB C-WAN Attachment Subnet** re-enters C-WAN **Hybrid Segment DXGW** for region (e.g. dxgw-use1) **Dual DX connections** (redundant, per edge connect) On-Prem S-DMZ **Fabric / CloudOne** (Fiserv Global Backbone) destination **Data Center** (USOMA01, USCHD01, UCHCZ01, DEFRA, APAC)

Key: The "send-via" policy means traffic must pass through Inspection before reaching the Hybrid segment. The DXGW is geo-bound per region — dxgw-use1 serves us-east-1, dxgw-use2 serves us-east-2, dxgw-usw2 serves us-west-2. No cross-geo leakage. Each DXGW has dual DX connections for redundancy at the edge connect location. Traffic traverses the Fabric/CloudOne backbone to reach the target DC.

North-South Inbound (On-Prem DC Workload e.g EC2)

On-Prem **Data Center** (e.g. USOMA01) **Fabric / CloudOne** backbone On-Prem S-DMZ **Dual DX connections** (redundant via Megaport / edge connect) **DXGW** for target region (e.g. dxgw-use1) C-WAN **Hybrid Segment** C-WAN policy: **send-via Inspection Segment Inspection VPC C-WAN Attachment Subnet GWLB** (GENEVE) **Palo Alto NVA** inspects inbound on-prem traffic returns to **GWLB C-WAN Attachment Subnet** re-enters C-WAN **Prod Segment Prod VPC EC2**

Key: Same inspection enforcement in reverse. On-prem traffic entering via the Hybrid segment **MUST** traverse Inspection before reaching any workload segment. The split DXGW design ensures traffic stays geo-local — US DCs use US DXGWs. Traffic originates from a DC, traverses the Fabric/CloudOne backbone, passes through the S-DMZ, and enters AWS via redundant DX connections.

M&A Flow (Acquisitions Account Workload e,g EC2)

Acquisition workload **Existing FiservMnA TGW Transit Gateway Attachment Route Table (CloudWAN)** TGW peering attachment C-WAN **MnA Segment** C-WAN policy routes to **Inspection Segment Inspection VPC C-WAN Attachment Subnet GWLB (GENEVE) Palo Alto NVA** inspects (M&A traffic is untrusted) returns to **GWLB C-WAN Attachment Subnet** re-enters C-WAN **Prod Segment Prod VPC EC2**

Key: M&A traffic enters Cloud WAN via the legacy TGW peering attachment into a dedicated MnA segment. This keeps acquired company traffic isolated in its own segment and always inspected before reaching production workloads. Two FiservMnA TGWs provide redundancy for the M&A integration path.

Shared Services (Any Segment Shared Svc VPC)

EC2 in any workload VPC C-WAN workload segment (Prod or Non-Prod) C-WAN **segment peering** (Prod Shared Svc, Non-Prod Shared Svc) **Shared Services Segment Shared Services VPC** R53 Resolver Endpoints, CloudWatch, S3 Log Buckets

Key: Shared Services peers directly with Prod and Non-Prod segments via C-WAN segment peering. This is the **only** cross-segment path that does NOT require Inspection traversal — because Shared Services only hosts infrastructure services (DNS, logging, monitoring), not application workloads.

Design Principles Summary

Principle	Implementation
Decentralized Ingress	Each workload VPC has its own IGW + ALB/WAF + GWLB. Per-region IGW provides the internet entry point.
Decentralized Egress	Each region has its own Regional NAT GW (in the Inspection VPC / Network Account). Egress exits through the regional NGW after inspection.
Centralized Inspection	ALL traffic (ingress, egress, E-W, N-S, M&A) must traverse the Inspection VPC where GWLB distributes to Palo Alto NVA targets via GENEVE encapsulation. Only exception: Shared Services peering.
Cloud WAN in ALL Paths	C-WAN segment policy (send-to / send-via) enforces that no traffic bypasses inspection. Segment isolation prevents direct cross-segment communication.
DXGW per Region	Each region has its own DXGW (dxgw-use1, dxgw-use2, dxgw-usw2) inside the Hybrid segment. No cross-geo leakage.
Dual DX per Edge Connect	Each DXGW has dual Direct Connect connections for redundancy at each edge connect location.
Fabric / CloudOne	On-prem traffic traverses the Fiserv Global Backbone / IBB SD-WAN overlay to reach the target Data Center (USOMA01, USCHD01, UCHCZ01, DEFRA, APAC).
M&A Isolation	Acquired companies enter via dedicated MnA segment through legacy FiservMnA TGW peering. Always inspected before reaching production.

6. Infrastructure as Code / Terraform

All lab infrastructure is managed via Terraform Enterprise with GitLab CI/CD runners.

Item	Details
GitLab Project-lab-repo	[GitLab repo URL]
State Backend	[S3, Dynamodb]

Sample Module Layout:

```
mcbb3-lab-terraform/modules/  
  cwan-core/      - C-WAN core network, segments, policies  
  cne/            - Cloud Network Edge per region  
  vpc/            - VPC, subnets, route tables  
  inspection-vpc/ - Inspection VPC with firewall subnets  
  network-firewall/ - AWS Network Firewall or NVA config  
  dxgw/           - Direct Connect Gateway per region  
  dx-connection/  - DX connections and VIFs  
  ibb-segment/    - IBB C-WAN segment and policies
```

7. Phased Deployment Runbook

The lab buildout is structured in sequential phases with dependencies and acceptance criteria mapped to Design Requirements (DRs).

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7.1 Phase 1: C-WAN Core Network & Segments

Dependencies: AWS account provisioned, Terraform Enterprise workspace configured, GitLab runner operational

DRs: DR 4.1, DR 2.10

- Deploy C-WAN core network with global policy document
- Create Prod, Non-Prod, Shared Services (peered to Prod + Non-Prod), and IBB segments across all three regions
- Deploy CNEs: CNE-US-East, CNE-APAC-Sydney, CNE-USWEST-Oregon
- Validate segment health and CNE state via console/API

Acceptance: All CNEs Available; four segments visible in all regions; Terraform state clean.

7.2 Phase 2: VPC Deployment & Intra-Region Routing

Dependencies: Phase 1 complete

DRs: DR 1.1, DR 2.1, DR 2.10

- us-east-1: Deploy Prod (10.20.0.0/21), Non-Prod (10.21.0.0/21), Inspection (10.22.0.0/22), Shared Services (TBD)
- ap-southeast-2: Deploy Prod (10.30.0.0/21), Inspection (10.32.0.0/22)
- us-west-2: Deploy Prod (TBD), Non-Prod (TBD), Inspection (TBD)
- Attach Prod VPCs to Prod segment, Non-Prod VPCs to Non-Prod segment, Shared Services to Shared Services segment
- Configure Shared Services segment peering to both Prod and Non-Prod segments
- Attach Inspection VPCs as separate C-WAN attachments
- Deploy EC2 test instances in each VPC for connectivity validation
- Validate segment isolation: Prod cannot reach Non-Prod directly
- Validate Shared Services reachable from both Prod and Non-Prod via peering

Acceptance: Routes in correct segments; Prod/Non-Prod isolated; Shared Services peered to both; Inspection attachments visible.

7.3 Phase 3: Cross-Region C-WAN Routing

Dependencies: Phase 2 complete

DRs: DR 1.2, DR 2.2, DR 2.9

- Validate C-WAN propagates VPC routes between CNEs across all three regions
- Test cross-region connectivity: us-east-1 ap-southeast-2, us-east-1 us-west-2, ap-southeast-2 us-west-2
- Verify route tables via API/CLI (aws networkmanager get-network-routes)
- Measure cross-region latency baselines
- Confirm VPC routes do NOT leak to DXGW attachments

Acceptance: Full mesh ping across all three regions; route tables correct; no DXGW leakage.

7.4 Phase 4: Split DXGW & Direct Connect Integration

Dependencies: Phase 3 complete, Megaport circuit delivered (target: March 20)

DRs: DR 1.3, DR 1.4, DR 2.3, DR 1.9, DR 1.10, DR 2.5

- Create three regional DXGWs: dxgw-us, dxgw-apac, dxgw-emea
- Attach DXGWs to their respective CNEs
- Provision Megaport VXC's as hosted DX connections
- Establish BGP peering over transit VIFs
- Validate geo-affinity and DXGW failover
- Confirm cross-DXGW isolation (DR 1.12)
- Test BGP community tagging on advertised routes (DR 2.5)

Acceptance: On-prem reachable via correct regional DXGW; no cross-geo DX leakage.

7.5 Phase 5: Inspection VPC & Service Chaining

Dependencies: Phase 2 complete (can parallel Phases 3-4)

DRs: DR 1.6, DR 4.8, DR 4.9

- Deploy firewall in each region's Inspection VPC (start with AWS Network Firewall)
- Configure C-WAN send-to/send-via actions to steer east-west and north-south traffic through Inspection VPC
- Validate firewall logs show inspected flows
- Test traffic blackhole when firewall removed (no bypass path)
- Optionally test third-party NVA (CheckPoint, Palo Alto) as alternative in one region

Acceptance: All east-west/north-south traffic through Inspection VPC; firewall logs confirm; no bypass.

7.6 Phase 6: IBB Architecture

Dependencies: Phase 4 complete

DRs: DR 1.5, DR 1.11, DR 1.12

- Deploy IBB C-WAN segment with dedicated route policies
- Test candidate IBB approaches:
 - **Option A:** C-WAN IBB segment with route policies restricting international traffic to approved paths
 - **Option B:** Accelerated Site-to-Site VPN tunnels for encrypted cross-region IBB underlay
 - **Option C:** Public VIF on existing DX for IBB overlay
- Validate international traffic paths and geopolitical routing constraints
- Document recommendation with pros/cons matrix

Acceptance: At least one IBB approach validated end-to-end with encryption; geopolitical constraints enforced.

7.7 Phase 7: Observability & Telemetry

Dependencies: Phase 3+ (can parallel Phases 4-6)

DRs: DR 3.1, DR 3.3, DR 3.4

Target State (Cloud-Agnostic):

- ThousandEyes — Network path visualization, latency monitoring across all CSPs
- Splunk — Centralized log aggregation for flow logs, firewall logs, C-WAN events
- Selector AI — AIOps anomaly detection and automated root cause analysis
- Wiz - Provides visibility into security posture in the cloud
- Forward Networks - Provide visibility on traffic pattern (traffic path analyzer)

Interim Lab (AWS Native):

- CloudWatch metrics for C-WAN (bytes in/out, packet counts per attachment)
- VPC Flow Logs for all lab VPCs (S3 + CloudWatch Logs)
- Network Manager route analytics and topology visualization
- CloudWatch alarms for DX connection state changes and BGP metrics - > <https://aws.amazon.com/blogs/networking-and-content-delivery/introducing-cloudwatch-metrics-for-aws-direct-connect-virtual-interface-bgp-health-and-prefix-count/>
- AWS Network Firewall logs (alert + flow) from Inspection VPCs - > <https://docs.aws.amazon.com/network-firewall/latest/developerguide/logging-monitoring.html> <https://docs.aws.amazon.com/network-firewall/latest/developerguide/firewall-logging.html>
- Document telemetry gaps and map to target state tool capabilities
- Lambda functions to push logs into S3 temporary then forwarded to the designated cloud agnostic observability/monitoring solution.

Acceptance: Interim dashboards operational; gaps documented with target state mapping.

7.8 Phase 8: End-to-End Validation & Stress Testing

Dependencies: All previous phases complete

DRs: DR 2.7, DR 2.8, DR 2.10

- Full traffic flow matrix: every region to every region, every region to on-prem, all through Inspection VPCs
- Blackout simulation: kill primary DX, measure convergence
- Brownout simulation: inject latency/packet loss, observe routing behavior
- Multi-tenancy validation: secondary segment isolation
- iperf bandwidth tests across C-WAN backbone and DX connections
- Compile final test results against DR matrix

Acceptance: All DRs validated or documented with gaps/workarounds; final report compiled.

8. Test Cases & Validation Matrix

Update Status as testing progresses. In Confluence, replace status text with /status macro for colored lozenges.

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Transport

DR	Description	Lab Test	Phase	Status
1.1	VPC-to-VPC same region	Ping Prod to Non-Prod in us-east-1	2	COMPLETED
1.2	VPC-to-VPC cross-region	Ping us-east-1 Prod to Sydney Prod	3	COMPLETED
1.3	VPC to on-prem (geo affinity)	US VPC reaches on-prem via dxgw-us only	4	NOT STARTED
1.4	VPC to on-prem (fallback)	Disable primary DX, verify alternate path	4	NOT STARTED
1.6	Network security enforcement	Traffic through Inspection VPC firewall	5	NOT STARTED
1.9	ECMP support	Multiple DX on same DXGW, verify load dist	4	NOT STARTED
1.12	Geopolitical routing avoidance	Verify APAC/EMEA traffic never routes via US DX	6	NOT STARTED

Routing

DR	Description	Lab Test	Phase	Status
2.1	Regional route propagation	VPC routes visible in regional CNE	2	COMPLETED
2.2	Cross-regional route propagation	VPC routes visible in remote CNEs	3	COMPLETED
2.5	BGP Path engineering	Verify DX-advertised routes traffic engineering	4	NOT STARTED
2.7	Sub-second convergence (blackout)	Kill DX link, measure convergence	8	NOT STARTED
2.9	End-to-end routing visibility	API/CLI route table dump all regions	3	NOT STARTED
2.10	Multi-tenancy (VRF-like)	Prod/Non-Prod segment isolation; no route leakage	2	COMPLETED

Observability

DR	Description	Lab Test	Phase	Status
3.1	Real-time flow telemetry	CloudWatch C-WAN metrics (interim); ThousandEyes (target)	7	IN-PROGRESS
3.3	Historical flow data	VPC Flow Logs to S3 (interim); Splunk (target)	7	IN-PROGRESS
3.4	Metrics push/pull /subscription	CloudWatch alarms (interim); Selector AI (target)	7	IN-PROGRESS

Security & Service Chaining

DR	Description	Lab Test	Phase	Status
4.8	AWS ANF/NVA integration	Route traffic through firewall in Inspection VPC	5	COMPLETED
4.9	Service chaining	C-WAN send-to/send-via steers traffic to Inspection VPC	5	COMPLETED
	Security DMZ plan guiding Rules	Cloud DMZ Perimeter Design	5	IN-PROGRESS